Status	Finished
Started	Friday, 9 October 2026, 10:35 PM
Completed	Friday, 9 October 2026, 10:38 PM
Duration	3 mins 22 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Binary search
c. Dynamic search
d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(log n) ✓
c. O👎
d. O(1)

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n-1
c. n ✓
d. n/2

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the left sublist
c. In the right sublist ✓
d. In the middle of the list

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By starting from the first element
c. By using a hash function
d. By dividing the list length by two ✓

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Bubble ✓

c. Complexity

d. Selection

Question **8**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The list is sorted
c. The search continues in the right sublist
d. The search ends successfully ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Discovering
c. Searching ✓
d. Mining

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Bubble search
c. Quick search
d. Merge search

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The list is sorted
d. The search continues from the beginning

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the search algorithm finishes
- c. When the list contains duplicate elements
- d. When the list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Hash search
- c. Binary search ✓
- d. Linear search

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list contains duplicate elements
- c. When the list is unsorted
- d. When the list is very small

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 90 and 94
- c. 89 and 94
- d. 89 and 99